Calculate the parameters

A. Query, Key and Value projections

Each projection transforms a 768-dimensional vector into another 768-dimensional vector.

The parameter formula for one projection is:

$$768 \times 768 + 768 = 590{,}592$$

The additional 768 represents the bias parameters.

There are three projections: Query, Key and Value.

$$3 \times 590{,}592 = 1{,}771{,}776$$

B. Attention output projection

$$768 \times 768 + 768 = 590{,}592$$

C. First feed-forward layer

This layer transforms the hidden size of 768 into the intermediate size of 3072.

$$768 \times 3072 + 3072 = 2{,}362{,}368$$

D. Second feed-forward layer

This layer transforms 3072 back to 768.

$$3072 \times 768 + 768 = 2{,}360{,}064$$

E. Two LayerNorm operations

Each LayerNorm has 768 scale parameters and 768 bias parameters.

$$2 \times (768+768) = 3{,}072$$

Final Calculation

Component	Parameters
Query, Key and Value projections	1,771,776
Attention output projection	590,592
First feed-forward layer	2,362,368
Second feed-forward layer	2,360,064
Two LayerNorm operations	3,072
Total for one encoder layer	**7,087,872**

Therefore, one BERT-base encoder layer contains **7,087,872 trainable parameters**, using the standard configuration and counting its linear-layer biases and LayerNorm parameters.

This excludes the embeddings, pooler, and classification head.